

# CO-2 → Propositional Logic

$P \rightarrow \sim P$  \* A ~~statement~~ or sentence is usually a collection of words.  
 $T \rightarrow F$   
 $F \rightarrow T$  \* A proposition is a declarative sentence for which we can assign truth values (T or F) but not both.

| Operator Name              | Symbol            | General usage  |
|----------------------------|-------------------|----------------|
| Conjunction                | $\wedge$          | and *          |
| Disjunction                | $\vee$            | or +           |
| Negation                   | $\sim$ or $\neg$  | not            |
| Conditional or implication | $\rightarrow$     | If then        |
| Bi-conditional             | $\leftrightarrow$ | If and only if |

→ One or more statements are connected with logical operators and the resultant statement is known as compound stmt.

$(\forall x, y) \rightarrow (x \rightarrow y) \wedge (y \rightarrow x)$

## Truth Tables

| ① | P | $\sim P$ | ② | P | q | $P \wedge q$ | ③ | P | q | $P \vee q$ |
|---|---|----------|---|---|---|--------------|---|---|---|------------|
|   | T | F        |   | T | T | T            |   | T | T | T          |
|   | T | F        |   | T | F | F            |   | T | F | T          |
|   | F | T        |   | F | T | F            |   | F | T | T          |
|   | F | T        |   | F | F | F            |   | F | F | F          |

| ④ Conditional ( $\rightarrow$ ) |   |                   | ⑤ Bi-conditional ( $\leftrightarrow$ ) |   |                       |
|---------------------------------|---|-------------------|----------------------------------------|---|-----------------------|
| P                               | q | $P \rightarrow q$ | P                                      | q | $P \leftrightarrow q$ |
| T                               | T | T                 | T                                      | T | T                     |
| T                               | F | F                 | T                                      | F | F                     |
| F                               | T | T                 | F                                      | T | F                     |
| F                               | F | T                 | F                                      | F | T                     |

### Session 9

Q. Identify which of the following are sentences and propositions.

- ① New Delhi is the capital of India. (Proposition)
- ② For  $x=2$ ,  $x+7=5$  (Proposition)
- ③ Did you understand? (Sentence)
- ④ What a beautiful morning (Sentence)
- ⑤  $x+3=5$  (Sentence) —
- ⑥ I'm a liar (Proposition)

Q. If p and q be the propositions

p: I bought a lottery ticket this week

q: I won the million dollar jackpot.

- ①  $\neg p$
- ②  $p \vee q$
- ③  $p \rightarrow q$
- ④  $\neg q$
- ⑤  $p \leftrightarrow q$
- ⑥  $\neg p \rightarrow \neg q$
- ⑦  $\neg p \wedge \neg q$
- ⑧  $\neg p \vee (p \wedge q)$

①  $\neg p$ : I didn't bring a lottery ticket this week.

②  $p \vee q$ : I bought a lottery ticket this week or I won the million-dollar jackpot.

③  $p \rightarrow q$ : If ~~that~~ I bought a lottery ticket this week then I won the million-dollar jackpot.

④  $p \wedge q$ : I bought a lottery ticket this week and I won the million-dollar jackpot.

⑤  $p \leftrightarrow q$ : I bought a lottery ticket this week if and only if I won the million-dollar jackpot.

⑥  $\neg p \rightarrow \neg q$ : If I didn't buy a lottery ticket this week then I didn't win the million-dollar jackpot.

⑦  $\neg p \wedge \neg q$ : I didn't buy the lottery ticket this week and I didn't win the million-dollar jackpot.

⑧  $\neg p \vee (p \wedge q)$ : I ~~didn't~~ ~~bought~~ a lottery ticket this week or I bought a lottery ticket and I won the million-dollar jackpot.



Write the following statements in symbolic form where  $p$  &  $q$  is.

$P$ : The computer is on

$q$ : The screen is working.

- ① The computer is on and the screen is working  $\rightarrow (P \wedge q)$
- ② The computer is on or the screen is working  $\rightarrow (P \vee q)$
- ③ If the computer is on then the screen is working.  $\rightarrow (P \rightarrow q)$
- ④ The computer is not on  $\rightarrow (\neg P)$
- ⑤ The screen is not working  $\rightarrow (\neg q)$
- ⑥ If the computer is off then the screen is not working  $(\neg P \rightarrow \neg q)$
- ⑦ The computer is on if and only if the screen is working  $(P \leftrightarrow q)$



Q. Explain the difference between Tautology, contradiction & contingency with logical reasoning. 03/09/24

A) Tautology  $\rightarrow$  A compound proposition is always true.

Contradiction  $\rightarrow$  A compound proposition is always false.

Contingency  $\rightarrow$  A proposition i.e; neither true nor false.

Tautology Ex:-

| P | $\sim P$ | $P \vee \sim P$ |
|---|----------|-----------------|
| T | F        | T               |
| F | T        | T               |

Contradiction Ex:-

| P | $\sim P$ | $P \wedge \sim P$ |
|---|----------|-------------------|
| T | F        | F                 |
| F | T        | F                 |

Contingency Ex:-

| P | Q | $P \vee Q$ |
|---|---|------------|
| T | T | T          |
| T | F | T          |
| F | T | T          |
| F | F | F          |

$\rightarrow$  Consider the truth tables for the following & verify whether it's tautology, contradiction or contingency.

i) if P then P or Q

ii)  $(P \rightarrow Q) \wedge (Q \rightarrow P)$

iii)  $(P \rightarrow Q) \wedge (P \wedge \neg Q)$

1)  $\exists P$  then  $(P \vee Q) \rightarrow P \rightarrow (P \vee Q)$

| P | Q | $P \vee Q$ | $P \rightarrow (P \vee Q)$ |
|---|---|------------|----------------------------|
| T | T | T          | T                          |
| T | F | T          | T                          |
| F | T | T          | T                          |
| F | F | F          | T                          |

$\therefore$  It is a tautology.

ii)  $(P \rightarrow Q) \wedge (Q \rightarrow P)$

| P | Q | $P \rightarrow Q$ | $Q \rightarrow P$ | $(P \rightarrow Q) \wedge (Q \rightarrow P)$ |
|---|---|-------------------|-------------------|----------------------------------------------|
| T | T | T                 | T                 | T                                            |
| T | F | F                 | T                 | F                                            |
| F | T | T                 | F                 | F                                            |
| T | T | T                 | T                 | T                                            |

$\therefore$  Contingency.

iii)  $(P \rightarrow Q) \wedge (P \wedge \neg Q)$

| P | Q | $(P \rightarrow Q)$ | $\neg Q$ | $(P \wedge \neg Q)$ | $(P \rightarrow Q) \wedge (P \wedge \neg Q)$ |
|---|---|---------------------|----------|---------------------|----------------------------------------------|
| T | T | T                   | F        | F                   | F                                            |
| T | F | F                   | T        | T                   | F                                            |
| F | T | T                   | F        | F                   | F                                            |
| F | F | T                   | T        | F                   | F                                            |

$\therefore$  Contradiction.

Q. Let P and Q be two propositions, consider the following two formulae in propositional logic.

$$S_1: (\neg P \wedge (P \vee Q)) \rightarrow Q$$

$$S_2: Q \rightarrow (\neg P \wedge (P \vee Q))$$

Verify whether  $S_1, S_2$  are tautologies or not.

$S_1$

| P | Q | $\neg P$ | $P \vee Q$ | $(\neg P \wedge (P \vee Q))$ | $(\neg P \wedge (P \vee Q)) \rightarrow Q$ |
|---|---|----------|------------|------------------------------|--------------------------------------------|
| T | T | F        | T          | F                            | T                                          |
| T | F | F        | T          | F                            | T                                          |
| F | T | T        | T          | T                            | T                                          |
| F | F | T        | F          | F                            | T                                          |

$\therefore$  Tautology.

$S_2$

| P | Q | $\neg P$ | $P \vee Q$ | $(\neg P \wedge (P \vee Q))$ | $Q \rightarrow (\neg P \wedge (P \vee Q))$ |
|---|---|----------|------------|------------------------------|--------------------------------------------|
| T | T | F        | T          | F                            | F                                          |
| T | F | F        | T          | F                            | T                                          |
| F | T | T        | T          | T                            | T                                          |
| F | F | T        | F          | F                            | T                                          |

$\therefore$  Contingency.

Logically equivalent

$\rightarrow$  If two compound statements are logically equivalent if and only if the truth values are same.



Q. Verify the following are logically equivalent or not.

i)  $P \rightarrow Q$  and  $\neg Q \rightarrow \neg P$

| P | Q | $P \rightarrow Q$ | $\neg Q$ | $\neg P$ | $\neg Q \rightarrow \neg P$ |
|---|---|-------------------|----------|----------|-----------------------------|
| T | T | T                 | F        | F        | T                           |
| T | F | F                 | T        | F        | F                           |
| F | T | T                 | F        | T        | T                           |
| F | F | T                 | T        | T        | T                           |

$\therefore$  Logically equivalent

Q)  $Q \rightarrow P$  and  $(\neg P) \rightarrow \neg Q$

| P | Q | $Q \rightarrow P$ | $\neg P$ | $\neg Q$ | $(\neg P) \rightarrow \neg Q$ |
|---|---|-------------------|----------|----------|-------------------------------|
| T | T | T                 | F        | F        | T                             |
| T | F | F                 | F        | T        | T                             |
| F | T | T                 | T        | F        | F                             |
| F | F | T                 | T        | T        | T                             |

$\therefore$  Logically equivalent.

### Session-10

Explain the converse, inverse and contrapositive of a conditional statement. Use a specific example to illustrate each concept.

A) conditional ( $P \rightarrow Q$ ) If 'P' and Q are any two propositions then the statement  $P \rightarrow Q$  is called conditional statement.

\* P is called the hypothesis or the premise and Q is called the consequence or conclusion.

① Given conditional statement is  $P \rightarrow Q$ ,  
converse is  $(Q \rightarrow P)$

Inverse is  $(\neg P \rightarrow \neg Q)$

Contrapositive is  $(\neg Q \rightarrow \neg P)$

\* If a real number  $x^2 > 0$  then  $x \neq 0$ .  
(Conditional statement  $P \rightarrow Q$ )

\* Converse  $(Q \rightarrow P)$

If a real number  $x \neq 0$  then  $x^2 > 0$

\* Inverse  $(\neg P \rightarrow \neg Q)$

If a real number  $x^2$  not greater than 0 then  $x = 0$ .

\* Contrapositive  $(\neg Q \rightarrow \neg P)$

If a real number  $x = 0$  then  $x^2 \neq 0$ .

② Given the conditional statement  $P \rightarrow Q$  where  
the 'P': is the computer is running  
Q: The software is installed, perform  
the following task.

i) Construct converse, inverse & contrapositive  
of the statement  $(P \rightarrow Q)$ .

ii) Determine whether the converse, inverse and  
contrapositive are logically equivalent to the  
statement  $(P \rightarrow Q)$  and provide truth  
tables for each to support your analysis.

A)

| P | Q | $P \rightarrow Q$ |
|---|---|-------------------|
| T | T | T                 |
| T | F | F                 |
| F | T | T                 |
| F | F | T                 |

Conditional statement  $(P \rightarrow Q)$ : If the Computer is running then the software is installed.

Converse  $(Q \rightarrow P)$ : - If the software is installed then the Computer is running.

Inverse  $(\neg P \rightarrow \neg Q)$ : - If the computer is not running then the software is not installed.

Contrapositive  $(\neg Q \rightarrow \neg P)$ : - If the software is not installed then the computer is not running.

| P | Q | $P \rightarrow Q$ | $Q \rightarrow P$ | $(\neg P \rightarrow \neg Q)$ | $(\neg Q \rightarrow \neg P)$ |
|---|---|-------------------|-------------------|-------------------------------|-------------------------------|
| T | T | T                 | T                 | T                             | T                             |
| T | F | F                 | F                 | F                             | F                             |
| F | T | T                 | T                 | T                             | F                             |
| F | F | T                 | F                 | T                             | T                             |

Q. Verify that  $\neg(P \vee Q)$  and  $\neg P \wedge \neg Q$  are equivalent.

| P | Q | $P \vee Q$ | $\neg(P \vee Q)$ |
|---|---|------------|------------------|
| T | T | T          | F                |
| T | F | T          | F                |
| F | T | T          | F                |
| F | F | F          | T                |



| $P$ | $Q$ | $\neg P$ | $\neg Q$ | $\neg P \wedge \neg Q$ |
|-----|-----|----------|----------|------------------------|
| T   | T   | F        | F        | F                      |
| T   | F   | F        | T        | F                      |
| F   | T   | T        | F        | F                      |
| F   | F   | T        | T        | T                      |

$\therefore$  Logically equivalent

Q. Verify that given the statements  $P \rightarrow (Q \wedge R)$ ,  $(P \rightarrow Q) \wedge (P \rightarrow R)$  Use logical equivalences to show, these two statements are

| $P$ | $Q$ | $R$ | $Q \wedge R$ | $P \rightarrow (Q \wedge R)$ |
|-----|-----|-----|--------------|------------------------------|
| T   | T   | T   | T            | T                            |
| T   | T   | F   | F            | F                            |
| T   | F   | T   | F            | F                            |
| T   | F   | F   | F            | T                            |
| F   | T   | T   | T            | T                            |
| F   | T   | F   | F            | T                            |
| F   | F   | T   | F            | T                            |
| F   | F   | F   | F            | T                            |

| $P$ | $Q$ | $R$ | $P \rightarrow Q$ | $P \rightarrow R$ | $(P \rightarrow Q) \wedge (P \rightarrow R)$ |
|-----|-----|-----|-------------------|-------------------|----------------------------------------------|
| T   | T   | T   | T                 | T                 | T                                            |
| T   | T   | F   | T                 | F                 | F                                            |
| T   | F   | T   | F                 | T                 | F                                            |
| F   | T   | T   | T                 | T                 | T                                            |
| T   | F   | F   | F                 | T                 | F                                            |
| F   | T   | F   | T                 | T                 | T                                            |
| F   | F   | T   | T                 | T                 | T                                            |
| F   | F   | F   | T                 | T                 | T                                            |

$\therefore$  Logically equivalent



Q.  $\neg(P \rightarrow Q)$  and  $P \wedge \neg Q$  analyse and state the logical equivalence between two statements using truth tables.

| P | Q | $P \rightarrow Q$ | $\neg(P \rightarrow Q)$ |
|---|---|-------------------|-------------------------|
| T | T | T                 | F                       |
| T | F | F                 | T                       |
| F | T | T                 | F                       |
| F | F | T                 | F                       |

| P | Q | $\neg Q$ | $P \wedge \neg Q$ |
|---|---|----------|-------------------|
| T | T | F        | F                 |
| T | F | T        | T                 |
| F | T | F        | F                 |
| F | F | T        | F                 |

$\therefore$  Logically equivalent.

Q. verify that  $(P \rightarrow Q) \vee (P \rightarrow R)$  and  $P \rightarrow (Q \vee R)$  logically equivalent.

Sol:-

| P            | Q            | R            | $P \rightarrow Q$ | $P \rightarrow R$ | $(P \rightarrow Q) \vee (P \rightarrow R)$ |
|--------------|--------------|--------------|-------------------|-------------------|--------------------------------------------|
| T            | T            | T            | T                 | T                 | T                                          |
| T            | F            | T            | F                 | T                 | T                                          |
| F            | T            | T            | T                 | T                 | T                                          |
| <del>T</del> | <del>F</del> | <del>F</del> | <del>T</del>      | <del>F</del>      | <del>T</del>                               |
| T            | F            | F            | F                 | F                 | F                                          |
| F            | T            | F            | T                 | T                 | T                                          |
| F            | F            | T            | T                 | T                 | T                                          |
| F            | F            | F            | T                 | T                 | T                                          |

| P            | Q            | R            | Q ∨ R        |
|--------------|--------------|--------------|--------------|
| T            | T            | T            | T            |
| T            | F            | T            | T            |
| F            | T            | T            | T            |
| F            | T            | F            | T            |
| <del>F</del> | <del>F</del> | <del>T</del> | <del>F</del> |
| T            | F            | F            | F            |
| F            | T            | F            | T            |
| F            | F            | T            | T            |
| F            | F            | F            | F            |

∴ Logically not equivalent

- ① Explain the universal quantifier and existential quantifiers. Illustrate your explanation with examples.

A) Predicate: - A part of declarative sentence describing the properties of object.

Ex:- Suman is a bachelor.  
Sunny is a bachelor.

In both the statements is a bachelor is called predicate.

\* Both Suman & Sunny have the same property of being a bachelor.

Let:  $(x)$  is a bachelor  
↓  
predicate variable.

Quantifiers: - These are the words that are referred to quantities such as some, all, there is, there exists...

Universal quantifier ( $\forall$ )

(for all, for every, for any, for each) considered as universal quantifiers.

→  $\forall x P(x)$  = For every element 'x' in the domain the Predicate  $P(x)$  holds true.  
(for all 'x'  $P(x)$ )

Ex:- 'All humans are mortal.' can be written as  $P(x) : x$  is a mortal.

## Existential quantifier ( $\exists$ )

(There is, There exists, some, at least one)  
Existential quantifiers.

Ex: There exists 'x' such that  $P(x): x^2 = 5$

$$\rightarrow \exists x P(x).$$

② Translate the system specifications into English with the predicate  $S(x, y)$ :  $x$  is in  $y$  and where the domain for  $x$  and  $y$  is of all systems and all possible states.  
 $x$ : all systems.

$y$ : all possible states.

A)  $\rightarrow \exists x S(x, \text{open})$

= There exists  $x$  is in state open.

= There exists all systems are in state open.

$$\rightarrow \forall x (S(x, \text{malfunctioning}) \vee S(x, \text{diagnostic}))$$

for all  $x$  is in state malfunctioning or  $x$  is in state diagnostic.

All systems are in state malfunctioning or diagnostic.

$$\rightarrow \exists x S(x, \text{open}) \vee \exists x S(x, \text{diagnostic})$$

there is  $x$  is in state open or there is  $x$  is in state diagnostic.



There is a system is in open state ~~ago~~  
there is a system is in diagnostic state.

→  $\exists x \neg S(x, \text{available})$

There exists  $x$  is in state unavailable

→  $\forall x \neg S(x, \text{working})$   $S(x, y)$   
for all  $x$  is in a state of not working

for all systems are not in a state of working.

③ Translate these specifications with  
 $F(P)$ : Printer 'P' is out of service,  
 $B(P)$ : Printer 'P' is busy.  
 $L(j)$ : Printer job is lost.  
 $Q(j)$ : Printer job is queued.

A) →  $\exists P (F(P) \wedge B(P)) \rightarrow \exists j L(j)$

If there exists printer 'P' is out of service  
and printer 'P' is busy then there exists  
printer job is lost.

→  $\forall P B(P) \rightarrow \exists j Q(j)$

If for all printers 'P' are busy then  
there exists printer job is queued.

$$\rightarrow \exists j (Q(j) \wedge L(j)) \rightarrow \exists p (F(p))$$

If there exists print job is queued and printer job is lost then there exists printer 'p' is out of service.

$$\rightarrow (\forall p B(p) \wedge \forall j Q(j)) \rightarrow \exists j L(j)$$

If for all printer 'p' is busy and for all printer job is queued then there exists print job is lost.

④ Translate into English the statement

$\forall x \forall y ((x > 0) \wedge (y < 0)) \rightarrow xy < 0$  where the domain for both variables consist of all real numbers.

A) For all real numbers 'x' and 'y' if  $x > 0$  and  $y < 0$  then their product  $xy$  is negative.

⑤ Translate the statement the sum of two positive integers is always positive into a logical expression.

A) Let 'x' and 'y' represent two integers. Let 'P' and predicate 'P' is a positive integer.

$P(z)$ : z is a positive integer.

$$\forall x \forall y (P(x) \wedge P(y) \rightarrow P(x+y))$$

⑥ Express each of these systems using predicate quantifiers and logical connectors.  
 → When there is less than 30 MB free on the hard disk a warning message is sent to all users. Convert into notation.

A)  $f(d)$ : There is less than 30 MB free bytes on the hard disk.  
 $w(u)$ : Warning message to users.  
 $f(d) \rightarrow \forall u w(u)$

⑦ No directories in the file system can be opened and no files can be closed when system errors have been detected.

A)  $D(d)$ : Directory  $d$  can be opened  
 $F(c)$ : files can be closed  
 $S(e)$ : system error are detected.  
 $S(e) \rightarrow (\forall d \neg D(d) \wedge \forall c \neg F(c))$

⑧ File system cannot be backed up if there is a user locked down.

$L(u)$ : user  $u$  is currently locked down.  
 $B(b)$ : File system can be backed up.  
 $\exists u L(u) \rightarrow \neg B(b)$



⑨ Video on demand can be delivered when there is a user ~~current~~ at least 8 mega bytes of memory available and the connection speed 56 kbps.

A) V: Video on demand can be delivered

M(m): At least 8 mega bytes of memory available.

S(s): Connection speed is 56 kbps.

$(\exists m M(m) \wedge S(s) \rightarrow V)$

Session-13

16/09/2

Direct proof ( $P \rightarrow Q$ )

\* When P is  $\alpha$  and  $\alpha$  is true where  $P \rightarrow \alpha$  is true.

Step 1:- Assume that P is true.

Step 2:- Prove that  $\alpha$  is true.

Indirect proof

① Contradiction

Ⓐ If only P is given

Step 1:- Assume that P is false.

Step 2:- Prove that P is true.

(Assumption is contradiction)

Ⓑ  $P \rightarrow Q$

Step 1:- Assume that P is true and Q is false.

Step 2:- Prove that P is false. Which is contradiction.



to our assumption  $q$  is false.

ii) Contrapositive ( $\neg q \rightarrow \neg p$ )

Step 1:- Assume  $q$  is false  $\rightarrow \neg q$  is true.

Step 2:- Prove that  $p$  is false which contradicts to our assumption  $q$  is false.

① IF 'n' is odd then  $n^2$  is odd

$P$ :  $n$  is odd

$q$ :  $n^2$  is odd

let  $p$  is true  $\Rightarrow n$  is odd.

$n = 2k+1$ ,  $k$  is an integer.

$$n^2 = (2k+1)^2$$

$$n^2 = 4k^2 + 4k + 1$$

$$n^2 = 2(2k^2 + 2k + 1)$$

$$n^2 = 2 \cdot n$$

So,  $n^2$  is also odd.

$\therefore q$  is true.

$$\begin{array}{c} p \rightarrow q \\ T \quad T \end{array} = T$$

② IF  $P \rightarrow Q$  is true.

Product of two odd integers is odd.

$P$ : Two integers are odd.

$q$ : Product is odd.

let  $m$  &  $n$  are two odd integers  $m = 2k_1+1$ ,

$$n = 2k_2+1$$

Assume  $p$  is true i.e.  $m$  &  $n$  are odd integers.

$$mn = (2k_1 + 1)(2k_2 + 1)$$

$$= (4k_1k_2 + 2k_2 + 2k_1 + 1)$$

$$= 2(2k_1k_2 + k_2 + k_1) + 1$$

$$= 2k + 1$$

$mn$  is odd.

→ Product is odd &  $q$  is true.

③ IF  $3n+2$  is odd then  $n$  is odd.

$P$ :  $3n+2$  is odd.

$Q$ :  $n$  is odd.

Contradiction:  $p$  is true; Assume  $q$  is false.

$n$  is even  $\Rightarrow n = 2k$

$$3n+2 = 3(2k) + 2$$

$$= 2(3k+1)$$

$$= 2 \times \text{integer} - \text{even}$$

$3n+2$  is even.

→  $P$  is false which is contradiction.

∴ Our assumption is wrong.

→  $q$  is true.

Hence,  $P \rightarrow q$  is true.

Contrapositive

$q$  is false  $\rightarrow \neg q$  is true

$\neg q \rightarrow \neg p$  is true.

1) Prove the following statements if  $n$  and  $m$  are both odd then  $n+m$  is even.

A) P:  $n$  &  $m$  are odd

Q:  $n+m$  is even

$n$  &  $m$  are odd

$$n = 2k_1 + 1 \text{ \& } m = 2k_2 + 1$$

$$n+m = 2k_1 + 1 + 2k_2 + 1$$

$$= 2k_1 + 2k_2 + 2$$

$$= 2(k_1 + k_2 + 1) = 2k$$

$n+m$  is even

$\therefore$  P is true;  $P \rightarrow Q$  is true.

2) P:  $n$  is odd &  $m$  is even  
Q:  $n+m$  is odd.

P is true.

$n$  is odd &  $m$  is even

$$n = 2k_1 + 1; m = 2k_2$$

$$= 2(k_1 + k_2) + 1$$

$$= 2k + 1$$

$n+m$  is odd.

$\Rightarrow$  Q is true

$P \rightarrow Q$  is true.



⑥  $n$  and  $m$  are even then  $n+m$  is even  
 $n+m$  is even.

$P$ :  $n$  and  $m$  are even.

$Q$ :  $n+m$  is even.

$P$  is true

$n$  &  $m$  are even

$$n = 2k_1 \quad m = 2k_2$$

$$n+m = 2k_1 + 2k_2$$

$$= 2(k_1 + k_2)$$

$$= 2k$$

$n+m$  is even.

$\Rightarrow Q$  is true

$P \rightarrow Q$  is true.

⑦  $P$ :  $n$  &  $m$  are both odd

$Q$ :  $nm$  is odd otherwise  $nm$  is even

$$n = 2k_1 + 1, \quad m = 2k_2 + 1$$

$$nm = (2k_1 + 1)(2k_2 + 1)$$

$$= 2(2k_1k_2 + k_1 + k_2) + 1$$

$$= 2k + 1$$

$nm$  is odd.

$Q$  is true.

$P \rightarrow Q$  is true.

## Session 11

① Prove that if  $m+p$  &  $n+p$  are even integers where  $m, n$  and  $p$  are integers. Then  $m+p$  is even. What kind of proof did you use?

A)  $P$ :  $m+n$  &  $n+p$  are even integers.  
 $Q$ :  $m+p$  is even.

Direct Proof:-

Let  $P$  is true

$m+n$  &  $n+p$  are even integers.

$$m+n = 2k_1$$

$$m = 2k_1 - n$$

$$n+p = 2k_2$$

$$p = 2k_2 - n$$

$$m+p = 2k_1 - n + 2k_2 - n$$

$$= 2(k_1 + k_2 - n)$$

$$= 2k, (k = k_1 + k_2 - n) \text{ an integer.}$$

$m+p$  is even.

$Q$  is true

$P \rightarrow Q$  is true.

② Use a proof by contradiction to show that if  $x+y \geq 2$  then  $x \geq 1$  or  $y \geq 1$  are real numbers then  $x \geq 1$  ~~and~~ <sup>or</sup>  $y \geq 1$ .

$$P: x+y \geq 2$$

$$Q: x \geq 1 \text{ or } y \geq 1$$

Original:  $P \rightarrow Q$

Contrapositive:  $\neg Q \rightarrow \neg P$

Assume  $Q$  is false.

$$x < 1 \text{ \& } y < 1$$

$$x = 1 - k_1 ; k_1 > 0$$

$$y = 1 - k_2 ; k_2 > 0$$

$$x + y = 1 - k_1 + 1 - k_2$$

$$= 2 - k_1 - k_2$$

$$= 2 - (k_1 + k_2)$$

$$\# x + y < 2$$

$\Rightarrow P$  is false.

$\therefore \neg Q \rightarrow \neg P$  is true.

③ Prove that if  $m$  and  $n$  are integers and  $mn$  is even then  $m$  is even or  $n$  is even.

A)  $P$ :  $mn$  is even

$Q$ :  $m$  is even or  $n$  is even.

Contradiction:-

$P$  is true and assume that  $Q$  is false.

$\Rightarrow m$  is odd and  $n$  is odd

$$m = 2k_1 + 1 \quad n = 2k_2 + 1$$

$$mn = (2k_1 + 1)(2k_2 + 1)$$

$$= 2(2k_1k_2 + k_1 + k_2) + 1$$

$$= 2k + 1$$

$mn$  is odd.

$\Rightarrow p$  is false which is contradiction.

$\therefore$  Our assumption is wrong.

$\Rightarrow q$  is true.

$p \rightarrow q$  is true.

Show that if ' $n$ ' is an integer and  $n^3+5$  is ~~odd~~ even, using contrapositive, contradiction.

A)  $p: n^3+5$  is odd.

$q: n$  is even.

original:  $p \rightarrow q$

contrapositive:  $\neg q \rightarrow \neg p$ .

\*If  $\neg q$  is false then  $\neg p$  is true.

' $n$ ' is odd

$$n = 2k+1$$

$$n^3+5 = (2k+1)^3+5$$

$$= 8k^3 + 12k^2 + 6k + 1 + 5$$

$$= 2(k^3 + 6k^2 + 3k + 3)$$

$$= 2k_1$$

$$\text{where } k_1 = k^3 + 6k^2 + 3k + 3$$

$n^3+5$  is even.

$p$  is false

$\neg q \rightarrow \neg p$  is true.

Contradiction: -  $p$  is true

Assume  $q$  is false

$n$  is odd

$$n = 2k+1$$



$$n^3 + 5 = (2k+1)^3 + 5$$

$$= 8k^3 + 12k^2 + 6k + 1 + 5$$

$$= 2(4k^3 + 6k^2 + 3k + 3)$$

$$= 2k_1$$

$$\text{where } (k_1 = 4k^3 + 6k^2 + 3k + 3)$$

$n^3 + 5$  is even

$\Rightarrow P$  is false which is contradiction.

$\therefore$  Our assumption is wrong

$\Rightarrow Q$  is true ; Hence  $P \rightarrow Q$  is true.